Sean Patten

1-934-500-4208 | spatten2@fordham.edu | LinkedIn | GitHub | Website

PUBLICATIONS

Patten, S., Chen, P.-Y., Schweikert, C., & Hsu, D.F. (2025). Enhancing Sentiment Classification with Machine Learning and Combinatorial Fusion. Proceedings of the 23rd IEEE International Conference on Pervasive Intelligence and Computing (PICom 2025). [Link] [Code]

Erdős Number: 2 (via Dr. D. F. Hsu)

RESEARCH & PROFESSIONAL EXPERIENCE

Research Intern, IBM
May 2023 - May 2024

- Led the research and development of a novel ensemble system for sentiment analysis, integrating models using combinatorial fusion.
- Achieved state-of-the-art (SOTA) accuracy of 97.072% on the IMDB dataset, surpassing existing benchmarks.
- Research resulted in a peer-reviewed publication (PICom 2025) and formed the basis of a Master's thesis.
- Collaborators: Dr. Pin-Yu Chen, Dr. D. F. Hsu, Dr. C. Schweikert.

Software Developer (Contract), Social Science Research Council
October 2024 - January 2025

- Developed a data collection and analysis pipeline to investigate trends in language model adoption and usage on the Hugging Face platform, supporting SSRC research initiatives.
- Work under direction of Tim O'Reilly and Ilan Strauss.

Consultant, Aspire Aviation
January 2025 - Present

- Developed predictive models for aircraft part failure and automated internal document generation systems using Python and LLMs.

PROJECTS

a: agent fleet & automation platform

https://github.com/seanpattencode/a

- Multi-device agent and automation platform run as daily infrastructure: compiled-C CLI, systemd-timer job hub, git-synced state across a fleet of machines, orchestrating multiple AI coding agents with automated review gates.

i: personal operator system

- Integrated one-keypress system for the full operator loop (health tracking, market attention, research capture, and document generation, including this resume); 40+ tools with token-budget-gated builds.

SuperCowsay: cowsay rewritten in assembly for maximum performance

https://github.com/seanpattencode/SuperCowsay

- 75x faster than the original Perl and 2.4x faster than a C implementation (measured with hyperfine); syscall count reduced from 676 to 2.

e: terminal text editor in C, faster than Torvalds's uemacs

https://github.com/seanpattencode/e

- Self-compiling single-file C editor (sh e.c builds it): 0.37 ms startup, 2.8x faster than Torvalds's uemacs of the same micro-emacs heritage; opens a 5,600-line file in 1.2 ms (nvim: 223 ms) and renders an interactive directory browser 3x faster than ls prints filenames (hyperfine-measured).

seanpatten.com: performance-minimal personal website

https://github.com/seanpattencode/seanpattencode.github.io

- Single ~5 KB file scoring 100/100 on PageSpeed: no images, external resources, or favicon request; native system fonts; a strict subset of the original 1991 HTML tag set, rendering correctly in everything from Lynx and 1993 Mosaic to current browsers.
- Hosts live two-player Pong, playable over the internet with no game server: WebRTC pairs players peer-to-peer at near-frame latency. Play at https://seanpatten.com/games/pong/
- Serves a real AI chat service: Gemma 4 running on a home machine (CPU only), replies streamed live to the browser. Chat at https://seanpatten.com/chat/
- Self-hosted dynamic pages with server state, served from a home machine through a Cloudflare tunnel by the a platform above.

Custom automated trading system

- In production with real capital; proprietary and cannot be shown or described in detail.

RESEARCH OBJECTIVE

Training, combining, and architecting ensemble strategies to improve performance, safety, and cost effectiveness.

EDUCATION

M.S. in Computer Science
August 2022 - May 2024

Fordham University Graduate School of Arts and Sciences, Lincoln Center, New York

- GPA: 3.667/4.0
- **Thesis:** Enhancing Sentiment Classification with Machine Learning and Combinatorial Fusion
- **Advisor:** Dr. D. F. Hsu

Relevant Coursework: Artificial Intelligence, Deep Learning, NLP, Machine Learning, Data and Information Fusion (Ensemble Learning)

B.S. in Computer Science & B.S. in Business Administration
August 2018 - May 2022

Fordham University, Gabelli School of Business, Bronx, New York

Relevant Coursework: Core Computer Science, Core Business Classes, Concentration in Entrepreneurship

AWARDS & HONORS

- National Merit Scholarship Semifinalist
- Johns Hopkins Center for Talented Youth Student
- We the People Competition 2018 New York State Winner & Nationals Finalist

TECHNICAL SKILLS

AI Agents: Claude Code, OpenAI Codex, Gemini CLI, Grok, Google Antigravity; used together in multi-agent workflows: worktree orchestration, review gates, MCP, browser automation, multi-device agent fleets, LLM fine-tuning, RAG

Languages & Frameworks: Python, C, C++, PyTorch, Tensorflow, Java, Javascript, SQL, CUDA, Assembly

Platforms & Tools: Kubernetes, Qiskit, Huggingface, Git

TEACHING

Club President, Fordham Digital Business Society *2021-2022*

Helped Fordham students learn practical digital skills. Held events on learning Python, personally teaching the basics to beginners, and increased attendance by over 1,000% compared to previous years.

QUANTUM COMPUTING

Xu, J., Wu, Y., Chen, S. Y.-C., Patten, S., Schweikert, C., & Hsu, D.F. (2026). Diversity-Aware Ensemble of Quantum-Classical Models via Combinatorial Fusion. Proceedings of the IEEE International Conference on Quantum Computing and Engineering (QCE 2026), Quantum Applications track, Toronto (accepted, not yet published).

Quantum Computing Researcher, Fujitsu Quantum Simulator Challenge *January 2026 - Present*

- First author of the Asia University team's accepted proposal (Q-Compress: Variational Entanglement Profiling for Extreme LLM Compression); currently running LLM weight-compression experiments on Fujitsu's 40-qubit tensor network simulator, a 1,024-node FX700 (A64FX) supercomputer cluster, via the QARP SDK and Qulacs.
- Maps transformer weight super-blocks onto qubit registers and uses entanglement profiling (Von Neumann entropy across bipartitions) to derive non-uniform, entropy-aware tensor-network compression topologies, targeting 80%+ memory reduction with under 2% accuracy loss against SVD-LLM, ASVD, SparseGPT, and CompactifAI baselines.

Research Team Lead, Laboratory of Informatics and Data Mining, Fordham *2026 - Present*

- Led the research team (Fordham LIDM students with Dr. Samuel Yen-Chi Chen, first author of the original QLSTM paper) building a reproducible quantum-classical Combinatorial Fusion Analysis pipeline: a 40-model pool of 26 quantum-circuit feature maps (including gradient-trained variational quantum classifiers) and 14 classical baselines, with cognitive-diversity-based model selection.
- Work produced the accepted paper Diversity-Aware Ensemble of Quantum-Classical Models via Combinatorial Fusion (QCE 2026).